

Worksheet: Determine Antoine constants for saturation pressure

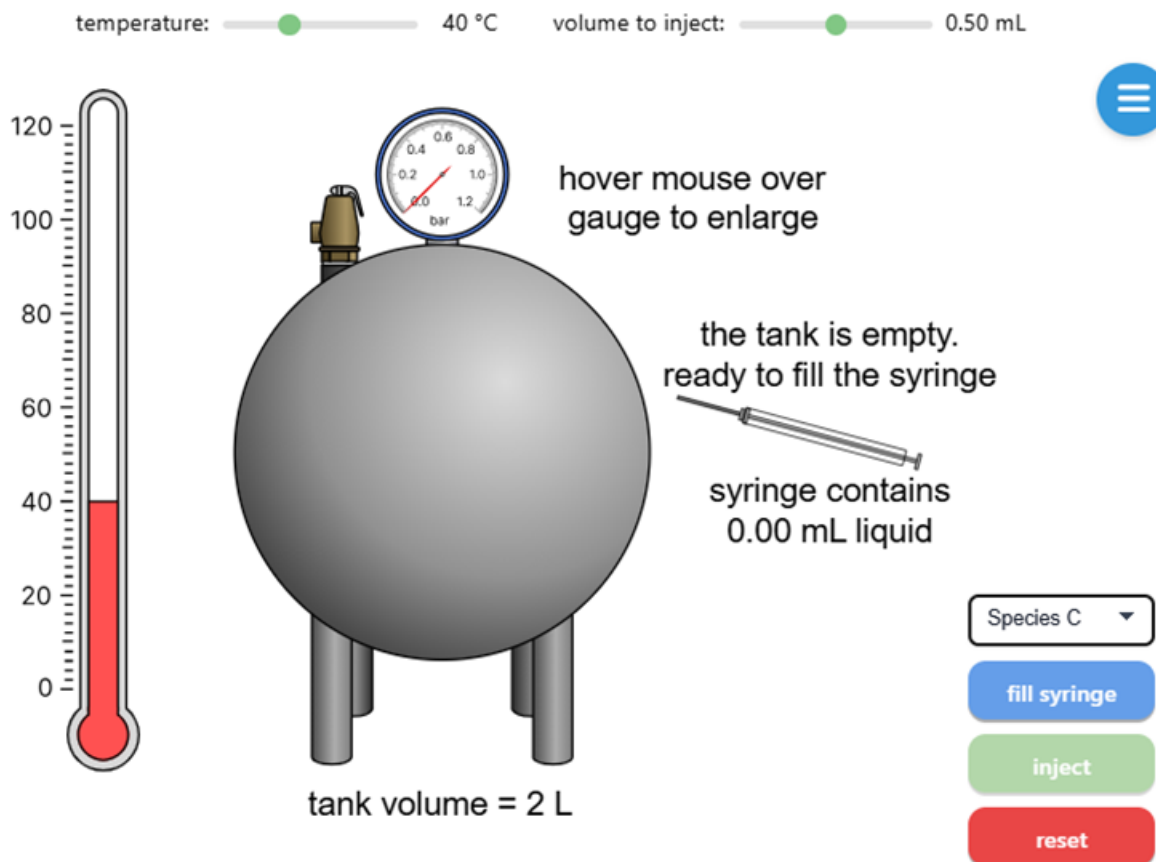
Name(s) _____

This experiment measures the saturation pressure (vapor pressure) of a single component over a range of temperatures by injecting enough liquid into an evacuated tank at constant temperature to obtain vapor-liquid equilibrium. The constants in the Antoine equation are determined from a series of temperature/pressure measurements. The heat of vaporization is obtained from the Clausius-Clapeyron equation.

Student learning objectives

1. Describe how saturation pressure changes with temperature.
2. Know how to measure saturation pressure.
3. Know how to fit data to a model to determine parameters.

Equipment



- A 2-L spherical tank in a constant-temperature bath. The tank has a port through which a syringe injects liquid. The pressure gauge on the tank reads absolute pressure. A pressure relief valve prevents pressures above the tank rating.
- A vacuum pump (represented by the reset button) evacuates the tank.
- A heater and temperature controller (represented by a temperature slider) adjust the tank temperature.
- A beaker of liquid (not shown) and a liquid syringe that can inject 1 mL or less of liquid.

Questions to answer before starting the experiment

1. This experiment measures saturation pressures by injecting liquid into an evacuated tank. Use the ideal gas law and the saturation pressure to explain why the amount of liquid needed to observe vapor-liquid equilibrium depends on temperature?
2. What must be present in the tank for the measured pressure to equal the saturation pressure: vapor only, liquid only, or both vapor and liquid? Explain.
3. Suppose enough liquid were injected at 50°C to measure saturation pressure. What happens to the pressure if more liquid were injected?
4. Given a limited temperature range for experiments, and given that you want to obtain the heat of vaporization from the Clausius-Clapeyron equation:

$$\log_{10} P^{sat} = -\frac{\Delta H_{vap}}{R} \frac{1}{T} + K$$
 where is P^{sat} in bar and the temperature is absolute temperature (K), how should the temperatures chosen for measurements be distributed in the range? That is, should the temperatures be equally spaced in the temperature range or in the inverse temperature range? Explain why.

Before starting

Evacuate the tank (with the reset button) so that the pressure gauge reads zero absolute pressure.

Select a molecule (A, B, C, D, E) to inject from the dropdown menu. Note that the temperature range of the slider depends on the molecule selected since Antoine constants only apply over a temperature range.

Molecule selected _____

Temperature range _____

Measure saturation pressures.

1. Pick the first temperature to make measurements.
2. Select the amount of liquid to inject.
3. Inject liquid with the syringe, allow the tank to equilibrate, and read pressure from the pressure gauge. Then inject more liquid until you are sure you are at vapor-liquid equilibrium. Record the temperature and the saturation pressure in the table below.

Temperature (°C)	Temperature (K)	1/T(K)	Saturation pressure (bar)

4. Change the temperature and repeat the procedure in step 3. Obtain measurements at 6 to 8 temperatures to obtain accurate Antoine constants. Select the temperatures to span the allowable range. For estimating heat of vaporization, temperature points should be distributed approximately evenly in 1/T(K).

Determine Antoine constants.

1. The Antoine equation is:

$$\log_{10} P^{sat} = A - \frac{B}{T + C}$$

Note that this is log base 10 and temperature is in °C.

2. Although Excel Solver can determine the Antoine parameters, it may have difficulty obtaining the best fit. Another approach is to use the LINEST function in Excel to apply multivariable linear regression to fit the saturation pressure (P^{sat} , bar) versus temperature (T, °C) data to a linearized form of the Antoine equation.

$$\log P^{sat} = A + (AC - B) \left(\frac{1}{T} \right) + (-C) \frac{\log P^{sat}}{T}$$

In this equation, the two independent variables (from your measurements) are $\left(\frac{1}{T} \right)$ and $\frac{\log P^{sat}}{T}$, and the dependent variable is $\log P^{sat}$ on the left side of the equation.

3. Values of the Antoine parameters (A, B, C) with their units obtained from a multivariable regression:

A = _____

B = _____

C = _____

4. Plot $\ln P^{sat}$ versus inverse absolute temperature and use the Clausius-Clapeyron equation to estimate the heat of vaporization (ΔH_{vap} , kJ/mol) of the molecule.

$$\ln(P^{sat}) = -\frac{\Delta H_{vap}}{R} \frac{1}{T} + K$$

where R is the ideal gas constant, P^{sat} is the saturation pressure at temperature T, and K is a constant. Note that the Clausius-Clapeyron equation uses the natural log.

$$\Delta H_{vap} = \text{_____} \text{ +/- } \text{_____}$$

5. Based on your fitted Antoine constants, what is the normal boiling point of the unknown liquid?

Questions to answer

1. After saturation is reached at one temperature, how does the split between liquid and vapor change if additional liquid is injected?
2. What are sources of error in the measurements?
3. Using literature data and your measurements of the heat of vaporization and the Antoine constants, which of these molecules is most likely your molecule?
benzene, toluene, methanol, ethanol, n-octane, n-heptane, water, iso-octane, cyclohexane
4. What safety measures would you employ if making this measurement in the laboratory?